

JISHIN BIJUMON GEORGE

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PROFESSIONAL SUMMARY

Electronics and Communication Engineering graduate focused on embedded systems and PCB design, with hands-on experience in microcontroller programming, sensor interfacing, schematic capture, PCB layout, hardware prototyping, debugging, and IoT development. Experienced with ESP32, ESP8266, Arduino, ATmega328P, and PIC16F877A, with practical use of KiCad, Altium Designer, MPLAB, and Proteus.

TECHNICAL SKILLS

- **PCB Design & EDA:** KiCad, Altium Designer, OrCAD, Schematic Capture, PCB Layout & Routing, ERC/DRC, Gerber Generation
- **IoT & Edge:** Firebase, Supabase, Edge Impulse
- **Microcontrollers:** ESP32, ESP8266, Arduino, ATmega328P, PIC16F877A
- **Embedded Development:** Embedded C, C, Python (Basic), Arduino IDE, MPLAB, Proteus, Sensor Interfacing, Hardware Testing
- **Interfaces:** UART, I²C, SPI, GPIO, ADC, PWM

EDUCATION

- B.Tech, Electronics & Communication Engineering — Amal Jyothi College of Engineering (Autonomous), Kanjirappally (KTU) — 2022–2026
- Higher Secondary Education (CBSE) — International Indian School, Dammam, Saudi Arabia — 2021–2022
- Secondary Education (CBSE) — International Indian School, Dammam, Saudi Arabia — 2019–2020

INTERNSHIP EXPERIENCE

Evolve Robotics, Ernakulam — Embedded Systems & Robotics Intern

Jul 2026 – Present

- Working on embedded systems and robotics projects — microcontroller programming, sensor interfacing, hardware testing, prototype development, and debugging — in a collaborative R&D environment.

NeST Digital, Ernakulam — IoT & Embedded Systems Immersion Program

Jun 2025

- Worked on IoT and embedded-system fundamentals including microcontroller programming and system integration; developed an ESP8266-based gas-leakage detection system with Firebase alerts.

Amal Jyothi College of Engineering — Embedded Systems Intern (Arduino IDE)

May–Jun 2024

- Worked on sensor interfacing and Arduino-based embedded C programming, gaining practical exposure to communication protocols and system-level integration.

EMBEDDED PROJECTS

BCI-Based Smart Wheelchair for Paralyzed Patients — EEG Signals & Line Following — ESP32 | Arduino | EEG/EOG | Firebase | IoT

- Developing a BCI-assisted wheelchair using EEG signals and eye-movement inputs for navigation with line-following and obstacle detection; integrated GPS tracking, health monitoring, and real-time telemetry.

Smart Specs for Visually Impaired People — ESP32-CAM & Object Detection (Published) — ESP32-CAM | Edge Impulse | IoT

- Developed a wearable assistive system for object/obstacle detection with sensor-based audio alerts; published as a research paper in 2025.

Human Elderly Health Monitoring System — ESP32 | MAX30102 | MPU6050 | DHT11 | GPS | GSM | Firebase

- Developed a wearable IoT system for health monitoring and fall detection using physiological, motion, environmental, and location sensors with emergency alerts.

PCB BOARDS

Custom PCB for BCI-Based Smart Wheelchair — Final Year Project — KiCad | 2-Layer PCB

- Designed, manufactured, and tested a custom PCB integrating power regulation, sensor integration, signal conditioning, and motor-control interfaces.

Custom Arduino Uno-Compatible Development Board — Ongoing — ATmega328P | KiCad

- Currently designing an ATmega328P-based development board with custom power, clock, reset, programming, and GPIO circuitry.

Custom ESP32-C3 Development Board — KiCad | 4-Layer PCB

- Designed an ESP32-C3-compatible development board with complete schematic capture and PCB layout.

Additional Projects: 5V–3.3V Regulator PCB, AC-to-DC Converter PCB, USB-to-UART Converter PCB, DR8833 Motor Driver PCB

KEY ACHIEVEMENTS

- Published a research paper on an ESP32-based assistive device for the visually impaired (MAT Journals Pvt. Ltd., 2025).
- Designed, built, and tested 6+ custom PCBs spanning power regulation, motor drivers, converters, and microcontroller dev boards.
- Completed 3 embedded-systems internships spanning robotics, IoT, and hardware prototyping.

CERTIFICATIONS & PUBLICATIONS

- Research Publication: "Smart Specs for Visually Impaired People using ESP32" — MAT Journals Pvt. Ltd., Vol. 2, Issue 2, 2025.
- Industry Immersion Program on Embedded Systems — NeST Digital, June 2025.
- Embedded Systems Internship (Arduino IDE) — Amal Jyothi College of Engineering (Autonomous), 2024.

REFERENCES

Agi Joseph George

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